

Solution Requirements

Date	28 September 2026
Team ID	6ab4d2876603573ccfb56348
Project Name	AI-WeatherWise
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system should allow registered users to securely log in and access the weather application.	High
2	Authorization levels	The system should provide appropriate access permissions for users and administrators based on their roles.	Medium

3	External interfaces	The system should integrate with a third-party Weather API to retrieve real-time weather information such as temperature, humidity, wind speed, and weather condition	High
4	Transactions processing	The system should process user weather requests, send them to the weather API, retrieve the required data, and return the processed weather information to the user.	High
5	Reporting	The system should provide weather information and AI-generated insights in a clear and understandable format for the requested location.	Medium
6	Business rules, etc.	The system should validate the location input, process valid weather data, and generate relevant AI-based recommendations based on the available weather conditions.	High
7	Compliance to laws or regulations	The system should protect user data and follow applicable data privacy and security requirements while handling user information and API data.	Medium
8	Other	The system should provide proper error handling, API response validation, and database connectivity to ensure smooth operation of the application.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should retrieve and process weather information within a reasonable response time.	The system should return weather information within 3 seconds under normal network conditions and should respond to API requests efficiently without unnecessary delays.
2	Scalability	The system should be capable of handling an increasing number of users and weather API requests without significant performance degradation.	System should support at least 100 concurrent users without major performance degradation.
3	Security & Data Privacy	The system should protect user data and API credentials using secure authentication, authorization, and appropriate data protection measures.	API keys and sensitive information must be stored securely in environment variables and not exposed in source code.
4	Reliability & Availability	The system should provide consistent weather information and continue to function reliably while handling API, database, or network failures gracefully.	System should successfully handle 95%+ of valid weather requests and provide proper error messages when the API is unavailable.
5	Usability & Accessibility	The system should provide a simple, user-friendly, and accessible interface that allows users to easily search and understand weather information.	Weather information should be displayed clearly, and users should be able to obtain weather results within 3 seconds under normal network conditions.

6	<i>Other</i>	The system should be maintainable, well-structured, and compatible with commonly used browsers and devices.	
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